

Drone-Remote Weapon Station

Details System Integration BOM

1. System-level BOM

BOM ID	Subsystem	Line item	Qty	Primary interfaces	Notes
CC-01	Computer & control	Jetson Orin NX control computer A	1	Eth, CAN, discrete	Active flight / payload control node
CC-02	Computer & control	Jetson Orin NX control computer B	1	Eth, CAN, discrete	Redundant peer for state sync and command cross-check
CC-03	Computer & control	State synchronization and heartbeat service	1	Eth	Software / middleware function spanning CC-01 and CC-02
SAF-01	Safety	Independent Safety MCU / ISM (Cortex-R class)	1	CAN, discrete, verified cmd bus	Final safety authority for torque enable, watchdog, limit enforcement
SAF-02	Safety	External watchdog module(s)	1 set	Discrete, CAN	Independent timeout supervision of control nodes
ARB-01	Safety / command arbitration	Dissimilar hardware arbiter (FPGA or equivalent)	1	Eth or verified command bus, discrete	Dual-compare and fail-safe command arbitration
ACT-01	Motion	CubeMars AK80-64 actuator - azimuth axis	1	Motor phases, encoder/CAN	Primary azimuth actuation
ACT-02	Motion	CubeMars AK80-64 actuator - elevation axis	1	Motor phases, encoder/CAN	Primary elevation actuation
ACT-03	Motion	CubeMars GL60-II gimbal motor	3	Motor phases, encoder	Weapon mechanics operation: Safe, Trigger, Dual Trigger
ACT-04	Motion	Gimbal MCU / servo-loop controller	1	CAN, discrete, motor control I/O	Closes motion loops and interfaces to motors / sensors
SNS-01	Sensors	Gimbal IMU	1	CAN / SPI / UART	Attitude and rate feedback

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SNS-02	Sensors	Angular rate gyro set	1 set	CAN / SPI	Rate stabilization reference
SNS-03	Sensors	Axis position encoders / resolver interfaces	1 set	Encoder lines / CAN	Position feedback for azimuth and elevation
SNS-04	Sensors	Pointing sensor A / homing / limit switches	1 set	Discrete / CAN	Travel limits and reference points
SNS-05	Sensors	Pointing sensor B/ homing / limit switches	1 set	Discrete / CAN	Travel limits and reference points
SNS-06	Sensors	Recoil Management Operation Verification A	1	Discrete / CAN	Recoil Management Healthy
SNS-07	Sensors	Recoil Management Operation Verification B	1	Discrete / CAN	Recoil Management Healthy
PAY-01	Payload	EO day camera	1	Eth or video, power	Primary visible-spectrum payload
PAY-02	Payload	Thermal camera	1	Eth or video, power	IR payload
PAY-03	Payload	Optional laser rangefinder / pointer	1	Discrete / serial	LRF payload
COM-01	Communications	Ground control station interface	1	RF / Eth	Remote operator control and display
COM-02	Communications	Primary RF link	1	RF, Eth	Primary remote link
COM-03	Communications	Independent backup RF link	1	RF, Eth	Backup command / telemetry path
COM-04	Communications	Diverse link manager	1	RF, Eth	Selects / manages primary and backup comms
LOG-01	Logging & test	Ruggedized data recorder / black box	1	Eth, HIL	Mission logging and safety evidence
PWR-01	Power	12 V battery pack A	2	Power	Primary energy source
PWR-02	Power	12 V battery pack B	1	Power	Redundant energy source

BOM ID	Subsystem	Line item	Qty	Primary interfaces	Notes
PWR-03	Power	Ideal diode OR-ing stage	2	Power	Battery isolation and hot redundancy
PWR-04	Power	Power controller	1	Power, discrete	Power conditioning and distribution
PWR-05	Power	Load shedding matrix	1	Power, discrete	Prioritized shedding under brownout / fault
PWR-06	Power	EMI filtering and isolated safety rails	1 set	Power	Clean power to safety and control domains
HMS-01	Health monitoring	HUMS controller	1	CAN / Eth, analog	Health and usage monitoring
HMS-02	Health monitoring	Temperature sensors	1 set	Analog / CAN	Thermal health
HMS-03	Health monitoring	Vibration sensors	1 set	Analog / CAN	Structural / motor vibration monitoring
HMS-04	Health monitoring	Power-rail quality monitor	1	Analog / CAN	Voltage / current / ripple health
MEC-01	Mechanical	Stabilized pan-tilt structure	1	Mechanical	Approx. 15 kg
MEC-02	Mechanical	Mechanical end stops and hard limits	1 set	Mechanical, discrete	Physical overtravel protection
MEC-03	Mechanical	Recoil Management System	1 set	Mechanical, discrete	Structural
MEC-04	Mechanical	Mounting frame, isolators, and damping hardware	1 set	Mechanical	Vehicle / aircraft integration and vibration attenuation

2. Connector and interconnect

The table below gives recommended connector families and interconnect practices for ruggedized platform. Exact shell size, insert arrangement, wire gauge, and environmental backshell selection should be finalized during the detailed harness design.

Interface	connector family	Typical use	Key notes
External power inlet / battery disconnect	MIL-DTL-38999 Series III or equivalent high-vibration circular	Battery pack, main power controller, service disconnect	Use keyed shells, rear grommet sealing, and current-rated contacts sized to peak inrush
External Ethernet / service port	38999 with Ethernet insert or ruggedized RJ45 only in protected service bay	Ground maintenance, recorder offload, HIL	Prefer sealed circular shell outside protected enclosure
Internal avionics low-mass I/O	Micro-D (MIL-DTL-83513)	Jetsons, ISM, recorder, watchdog, discrete I/O	Density for internal racks; strain-relieve harness at shell
CAN bus node interconnect	Micro-D, Nano-D, or sealed circular depending environment	Safety MCU, gimbal MCU, HUMS, sensors	Maintain bus continuity and controlled stub lengths
Motor power phases	High-current circular connector, 38999 power insert, or dedicated motor connector	AK80-64 and GL60-II motor phases	Separate from low-level sensor harnesses; add braid termination at enclosure entry
Encoder / sensor lines	Micro-D or shielded mini circular	Encoders, IMU, limit switches	Individually shielded twisted pairs for differential encoder channels
RF front-end	Qualified RF coax interface (e.g., TNC / SMA only in protected RF bay)	Primary and backup RF links	Select based on frequency plan, sealing, and cable loss budget
Safety kill / enable discrete	Dedicated locking circular or Micro-D discrete pins	Torque enable, watchdog reset, load shed, E-stop	Keep safety enable separate from non-critical payload discretes

3. Wiring harness

Harness identifiers below are integration-level baselines, intended to support routing, shielding, and connector-count estimation. Final wire gauges and pin assignments depend on detailed current draw, EMC testing, and packaging.

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Harness ID	Endpoints	Signals / media	Recommended construction	Comments
H-ETH-01	GCS <-> link manager	Primary IP / RF backhaul	Shielded Cat6A or quad-ax, ruggedized ends	Primary command and telemetry path
H-ETH-02	Backup RF <-> link manager	Backup IP / RF backhaul	Shielded Ethernet / coax bundle	Physically separate from H-ETH-01 where practical
H-ETH-03	Link manager <-> Jetson A/B	Ethernet control / state / video	Dual shielded Ethernet links	Can be direct or via compact switch backplane
H-ETH-04	Jetsons <-> recorder / HIL	Logging / test / maintenance	Shielded Ethernet	Support recorder and maintenance traffic
H-CAN-01	Safety MCU <-> gimbal MCU <-> HUMS <-> sensors	CAN bus	120 ohm terminated twisted pair, shielded	Short stubs to each node; terminate only at bus ends
H-DISC-01	Watchdogs <-> Jetson A/B <-> Safety MCU	Heartbeat, timeout, enable discretes	Shielded multi-core discrete harness	Critical timing and fail-safe lines
H-VER-01	Arbiter <-> Safety MCU <-> Gimbal MCU	Verified command bus / discrete handshakes	Shielded multi-pair harness	Command arbitration and safety gating
H-PWR-01	Battery A/B <-> diode OR stage <-> power controller	Main 12 V distribution	High-current pair with braid / foil shield as needed	Size to peak continuous and transient load
H-PWR-02	Power controller <-> load shedding matrix	Conditioned power rails	High-current multi-core	Feeds prioritised loads
H-PWR-03	Load shedding matrix <-> avionics	Protected avionics power	Individual branch circuits, labelled	Separate safety rails from payload / compute rails
H-MTR-01	Gimbal MCU / drivers <-> AK80-64 azimuth	Motor phases + feedback	High-current motor cable plus shielded feedback pair	Route apart from camera video
H-MTR-02	Gimbal MCU / drivers <-> AK80-64 elevation	Motor phases + feedback	High-current motor cable plus shielded feedback pair	Same EMC practice as H-MTR-01

Harness ID	Endpoints	Signals / media	Recommended construction	Comments
H-MTR-03	Gimbal MCU / drivers <-> GL60-II motors	Motor phases + feedback	Lightweight motor harness bundle	Breakout at moving joints with strain relief
H-SNS-01	Payload / IMU / encoders / limit switches <-> controllers	Sensor signals and low-current power	Individually shielded pairs where needed	Group by noise sensitivity; isolate gyro / IMU lines

4. Bus topology

Communications partitioning - Ethernet carries high-bandwidth traffic such as state synchronization, maintenance, logging, and payload data. CAN carries deterministic low-bandwidth control, health, and sensor traffic for the safety MCU, gimbal MCU, and distributed sensors. Safety-critical discrete lines remain separate from both Ethernet and CAN for watchdog, torque-enable, and load-shed functions.

Bus	Participants	Role	Integration notes
Ethernet / IP	GCS, link manager, Jetson A, Jetson B, recorder / HIL, payload network devices	Command, telemetry, logging, maintenance, video	Use physical separation between primary and backup RF ingress where possible
CAN	Safety MCU, gimbal MCU, HUMS, IMU, encoders / smart sensors	Deterministic control and health traffic	Maintain 120 ohm end termination and keep branch stubs short
Discrete I/O	Watchdogs, safety MCU, power controller, load shedding, limit switches	Fail-safe enables, watchdog timeout, reset, hard limits	Treat as safety paths; avoid multiplexing with non-critical payload controls
Power rails	Battery A/B, diode OR, power controller, load shedding, avionics loads	Energy distribution	Partition safety rails from high-noise payload and motor rails

5. BOM to block-diagram mapping

Architecture block	Mapped BOM IDs	Included items in this pack	Notes / exclusions
Ground control station	COM-01	Operator control endpoint	Only generic control/display function retained
Primary and backup links	COM-02, COM-03, COM-04	Primary RF, backup RF, diverse link manager	Weapon command semantics included
Redundant compute nodes	CC-01, CC-02, CC-03	Jetson A/B and state sync	Tracking / stabilization / payload management only
Safety and arbitration	SAF-01, SAF-02, ARB-01	ISM, watchdogs, command arbiter	Retained as motion-safety controls; fire interlock excluded
Motion control	ACT-01 to ACT-04	Actuators, gimbal motors, gimbal MCU	Azimuth / elevation / stabilization only
Payload and sensors	SNS-01 to SNS-04, PAY-01 to PAY-03	IMU, gyros, encoders, camera payloads, optional LRF	Weapon trigger, ballistics functions
Power domain	PWR-01 to PWR-06	Batteries, OR-ing, controller, load shedding, EMI / safety rails	Supports compute, safety, and payload loads
Health monitoring	HMS-01 to HMS-04	HUMS and environmental / electrical monitors	Used for maintenance and prognostics
Mechanical platform	MEC-01 to MEC-03	Structure, hard limits, mounts, damping	15 kg class platform

6. BOM to FMECA traceability

This section links each BOM group to representative failure modes so the integration package can feed directly into the FMECA & hazard log.

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BOM IDs	Primary failure mode	Local effect	System effect	Detection / mitigation
CC-01, CC-02, CC-03	State divergence or control-computer crash	Loss of one compute lane or command mismatch	Degraded control authority; transition to safe mode	Heartbeat supervision, dual compare, watchdog reset, safe-state fallback
SAF-01, SAF-02, ARB-01	Safety MCU failure or unsafe pass-through	Loss of safety gating	Potential unsafe motion if not contained	Independent watchdog, fail-off torque enable, arbiter cross-check
ACT-01, ACT-02	Azimuth / elevation actuator jam, runaway, or overcurrent	Axis fails to track or drives to stop	Pointing loss, structural overload risk	Current limit, encoder plausibility, mechanical stops, safe disable
ACT-03, ACT-04	Gimbal motor or servo-loop instability	Fine stabilization degraded or oscillatory	Image blur, pointing instability	Loop monitoring, thermal protection, gain scheduling / damping
SNS-01 to SNS-04	IMU / gyro / encoder drift or loss	Bad feedback data	Incorrect stabilization or limit overrun	Sensor voting / plausibility, timeout, hard limits
PAY-01 to PAY-03	Payload sensor loss or degraded video	Loss of mission sensor feed	Reduced situational awareness or ranging	BIT, link monitoring, graceful payload isolation
COM-01 to COM-04	Primary link loss or manager failure	Command / telemetry interruption	Fallback to backup link or local safe hold	Link manager switchover, comm-loss policy, recorder evidence
PWR-01 to PWR-06	Brownout, rail noise, OR-ing fault	Power interruption or degraded rail quality	Controller resets, sensor faults, loss of actuation	Load shedding, isolated safety rails, power-quality monitoring
HMS-01 to HMS-04	Health monitor blind or sensor failure	Missed prognostic data	Reduced maintenance insight; latent fault exposure	Cross-check against controller BIT and maintenance inspection

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BOM IDs	Primary failure mode	Local effect	System effect	Detection / mitigation
MEC-01 to MEC-03	Mount looseness, overtravel, damping degradation	Mechanical stress or excessive vibration	Reduced stabilization, hardware damage risk	Torque limits, inspections, vibration monitoring, hard stops

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7. Appendix A. Architecture Image

DUAL-REDUNDANT FAULT-TOLERANT AIRBORNE ARCHITECTURE

Defined in the 'System Architecture Report (SAR)' Dual Redundant Payload' and file. 'general architecture.pptx'

